Name: ..... Due: **Next Class / Friday**

C3 Mixers & DAWs HW 1 – Alanis Morissette 'You Oughta Know' & Alesis ADAT

Artist: Alanis Morissette (Jagged Little Pill, 1995, Producer: Glen Ballard)
https://www.youtube.com/watch?v=NPcyTyilmYY

1. "You Oughta Know" was tracked in Glen Ballard's home studio on Alesis ADAT digital tape. Identify the instrument arrangements and production methods below. (3)

Instrument / Track Element	Arrangement & Production Processing
Lead Vocal (Alanis Morissette)	Raw demo vocal take, high-pass filter, subtle 1176 compression
Bass Guitar (Flea)	Direct Injection (DI) blended with miked Gallien-Krueger amp
Drums (Taylor Hawkins / Matt Laug)	
Electric Rhythm Guitars (Dave Navarro)	

2. The Alesis ADAT recorded 8 tracks of uncompressed digital audio onto consumer S-VHS tape. What were the exact bit-depth and sampling rate specifications of the original ADAT? (2)

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3. Why was Modular Digital Multitrack (MDM) considered revolutionary for independent producers compared to traditional 2-inch analogue tape machines? (3)

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4. Explain the function of the High-Pass Filter (HPF) and the Input Pad (-20 dB) located at the top of a console channel strip during vocal and drum tracking. (4)

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5. Explain the difference between routing an external processor via a Channel Insert versus an Auxiliary Send. Give one practical studio example of each routing method. (4)

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6. In digital recording, explain the relationship between Bit Depth and Dynamic Range, and describe why engineers add Dither when truncating from 24-bit to 16-bit: (4)

Bit Depth & Theoretical Dynamic Range (6 dB Rule) (2)
Dynamic range formula (6 dB per bit) and comparison of 16-bit (96 dB) vs 24-bit (144 dB):
Dither Application & Quantization Distortion (2)
Why low-level random TPDF noise replaces correlated harmonic distortion during bit reduction:

Total:	**/18**